Alapan Das

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Employement

Laboratoire de Physique de l'École Normale Supérieure, ENS-PSL, Paris, Doctorat en physique *Sept 2025 – present*

Thesis title: Unusual phase transitions due to quantum measurements
Advisor: Prof. Adam Nahum

Education

Tata Institute of Fundamental Research, Mumbai *Aug 2022 – July 2025*
Master of Science in Theoretical Physics

- GPA: 8.4/10

Thesis title: Ising model on hyperbolic lattices (Grade: 9/10)
Advisor: Prof. Kedar Damle.

Visva Bharati University, Santiniketan *July 2019 – Aug 2022*
Bachelor of Science, Physics

- GPA: 8.55/10

Birbhum Zilla School, Suri *2009 – 2019*
10+2, Science

- Grade: 94%

Publications and pre-prints

Bayesian phase transition for the critical Ising model: Enlarged replica symmetry in the epsilon expansion and in 2D, Kay Jörg Wiese, **Alapan Das**, Adam Nahum ; arXiv:2604.23346 (*to be updated with version 2*)

Projects

Ising Model on Hyperbolic Lattices (Master's thesis) *Jan 2025-June 2025*

- Studied the free fermionic spectrum of the Ising model on the hyperbolic lattices. In this project, I constructed a dimer mapping for the hyperbolic lattice that respects the Fuchsian group symmetry. I employed the generalised Bloch theorem for non-Abelian Fuchsian groups to study the free fermionic spectrum of the system to understand several properties of this system. (*Manuscript under preparation, with Prof. Kedar Damle*)

Free Fermionic Representation of RBIM *Feb 2024-Dec 2024*

- We constructed a dimer mapping with a real Hamiltonian for random-bond Ising models in two dimensions. From this, we studied the low-energy (LE) modes of this free fermionic model, which are introduced by frustration. We connected the frustration to a monomer mixing problem to explain these LE modes and found a lower bound for the cumulative distribution function of them, which helps us understand Griffith's phase better. (*Manuscript under preparation, with Prof. Kedar Damle*)

Academic achievements

Joint Entrance Screening Test, Physics *Mar 2022*
Score-74/100, All India Rank - 1

IIT-IISc Joint Admission test for Master's, Physics *Feb 2022*
Score-89.67/100, All India Rank - 2

Jagadish Bose National Science Talent Search (JBNSTS), Kolkata *2019 & 2017*
Senior and Junior Scholar

DST-Inspire scholarship *2019-2022*
Govt. of India

Conferences and Workshops

School on Frustrated Quantum Magnets *Oct 28 - Nov 8, 2024*
École de Physique des Houches, Les Houches, France

Stability of Quantum Matters In and Out of Equilibrium at Various Scales *15-24 Jan, 2024*
ICTS-TIFR, Bangalore

IISER-JBNSTS workshop, IISER Kolkata *July 2018*
Workshop organised for JBNSTS junior scholars

Work Experience

Teaching assistant for Advanced Quantum Mechanics course *Aug-Dec 2024*
Instructor: Prof. Rajdeep Sensarma

Teaching assistant for Mathematical methods course *Aug-Dec 2023*
Instructor: Prof. Shravan Hanasoge

Skills

C++, Python, Mathematica, Inkscape

Extracurricular activities

Secretary of TIFR movie club *Aug 2024 - May 2025*
I was a co-secretary of the movie club of the TIFR student society